JARETT POLINER
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Research engineer at Los Alamos National Laboratory working at the intersection of generative AI, simulation, and complex physical systems. Develops learned physical representations, generative models, and multimodal world models from sparse observations, combining computational mechanics, active learning, and rigorous evaluation to advance simulation-ready environments and intelligent physical systems.

EXPERIENCE

Los Alamos National Laboratory, Earth and Environmental Sciences (LANL, EES) — Los Alamos, NM
Postdoctoral Research Associate — October 2025–Present

- Develop generative models of 3D material microstructures from partial micro-CT data, integrating graph-based modeling and latent diffusion to produce simulation-ready geometries; mentor graduate researchers in model development and evaluation.
- Develop recursive void-conditioned packing for multiscale granular super-resolution, incorporating non-overlap constraints and realistic particle-size and shape distributions.
- Validate generated microstructures against micro-CT ground truth and downstream physics simulations, assessing geometric fidelity and preservation of structure–property relationships.
- Contribute to Transparent Earth, a multimodal subsurface foundation model; investigate context scaling through query-conditioned observation selection and learned retrieval.

Urban Traffic Forecasting and World-Model Research — Columbia University / CMT
Collaborator — January 2026–Present

- Build reproducible data ingestion and evaluation pipelines for asynchronous traffic, camera, and event streams; design controlled ablations to quantify the residual predictive value of auxiliary observations.
- Develop a graph-based traffic-state estimation and forecasting model using sparse taxi telemetry, road-network structure, and optional camera and event context in a partially observed Manhattan network.
- Quantified the cost of missing observations: withholding 10–30 minutes of recent Manhattan probe history increased overall forecast error by 2.5–4.4%, reaching 9.8% at the 10-minute forecast horizon.
- Improved sparse-sensor forecasting robustness: at 75% sensor withholding, sparse-observation training reduced withheld-node error by 27% on METR-LA and 55% on PEMS-BAY.

Columbia University, Civil Engineering Department — New York, NY
Doctoral Research Assistant — September 2019–August 2025

- Developed a sample-efficient active-learning framework for expensive scientific simulations, combining Bayesian optimization, maximum-entropy sampling, and conditional latent diffusion to reconstruct high-dimensional response surfaces with fewer than 30 simulation queries.
- Replaced per-material-point nonlinear return-mapping solves with a learned projection in a mesh plasticity solver, yielding order-of-magnitude speedups on large-scale finite-element benchmarks (CMAME 2025, co-first author).
- Generated simulation-ready 3D geometries using graph-based models (GraphRNN and VGAE), conditional diffusion models, and flow-based generative models, producing microstructures whose topology and morphology match micro-CT references and feed directly into downstream solvers.

Air Force Research Laboratory, Munitions Directorate (AFRL, RW) — Eglin Air Force Base, FL
AFRL Scholar — June 2023–August 2023

- Applied generative AI, graph convolutional neural networks, and Fourier shape analysis to predict void shapes for prescribed critical threshold velocities during pore-collapse scenarios.
- Presented research outcomes to Air Force personnel and contributed to an internal research paper on data augmentation and generative prediction methods for energetic materials.

Columbia University, Civil Engineering Department — New York, NY
Graduate & Undergraduate Research Assistant — May 2018–December 2019

- Developed finite-element simulation workflows for pediatric heart-valve prosthetics and computational workflows for 3D concrete printing.

SELECTED PUBLICATIONS AND MANUSCRIPTS

Published and Accepted for Publication

- Poliner, J., Sun, W. C., Alshibli, K. A., & Regueiro, R. A. (2026). Deep generative graphs for synthesizing microstructure of topology-preserved polymer-bonded explosives. *Engineering with Computers, 42*, Article 207. DOI
- Poliner, J., Crone, J. C., Bryant, E. C., Bennett, K. C., & Sun, W. C. (2026). A generative framework for constructing strength models of structured materials from sparse data. *Scientific Reports. Accepted October 2026.*
- Xiao, M.*, Poliner, J.*, & Sun, W. C. (2025). Geometric learning for computational mechanics Part IV: Efficient mesh-based plasticity from a domain-specific foundation model. *Computer Methods in Applied Mechanics and Engineering, 446*, 118310. DOI

Manuscripts in Preparation

- Poliner, J., & Di, X. (2026). FleetNet: Learning urban traffic state dynamics under partial observability and disruptions. Manuscript in preparation.
- Doan, J.*, Poliner, J.*, Bryant, E. C., Bennett, K. C., Miller, N. A., & Hill, R. G. (2026). Variational graph autoencoding for latent compression and reconstruction of granular microstructures. *Manuscript in preparation*.
- Poliner, J., Bryant, E. C., Bennett, K. C., & Miller, N. A. (2026). Particle packing super-resolution: Recursive void-conditioned generation of multiscale granular microstructures. *Manuscript in preparation*.

*Co-first authors; contributed equally.

SKILLS

Machine Learning: PyTorch, PyTorch Geometric, Bayesian optimization, active learning, graph neural networks, transformers, diffusion models, flow matching, representation learning, multimodal systems, world models
Simulation & Evaluation: active experimental design, state estimation, controlled ablations, held-out validation, numerical simulation, model benchmarking
ML Systems & Infrastructure: Linux, Git, Docker, containerized execution, HPC clusters, GPU computing, CUDA, experiment orchestration, large-scale data pipelines
Scientific Computing: finite-element and discrete-element methods, numerical optimization, inverse problems, solver stability
Programming: Python, MATLAB
Simulation Tools: NVIDIA Warp, ABAQUS, Yade
Clearance: Active U.S. Department of Energy (DOE) Q Clearance

EDUCATION

Columbia University, Fu Foundation School of Engineering and Applied Science
Ph.D., *Civil Engineering with a concentration in Computational Mechanics* August 2025
Dissertation: *Generative Modeling with Sparse Data for Solid Mechanics*
Awards: 2023 AFRL Scholarship, 2022 DoD NDSEG Poster Award, 2020 NDSEG Fellowship, 2020 Guggenheim Fellowship

M.S., *Civil Engineering with a concentration in Computational Mechanics* May 2020

B.S., *Engineering Mechanics* May 2019
Minors: Applied Mathematics, Civil Engineering and East Asian Studies

Honors: 2018 & 2019 Dean's List, 2018 Concrete Institute Foundation (CIF) Scholarship

ATHLETICS AND LEADERSHIP

Columbia University, NCAA Division I Varsity Men's Fencing Team New York, NY
Epee Fencer (Captain, 2018–2019); 2016 and 2019 NCAA Team National Champions August 2015–2019